

A Unified Framework With Capped Tensor Norm Minimization for Multiview Subspace Learning

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Abstract—In recent years, multi-view subspace clustering (MVSC) has emerged as a powerful tool for high-dimensional data analysis, due to its ability to exploit intrinsic structures across multiple feature spaces. However, existing MVSC methods still suffer from two shortcomings: 1) the prevalent use of convex low-rank approximations inadequately exploits complementary subspace information, and 2) the neglect of intrinsic geometric relationships and the execution of spectral clustering as a separate step often lead to suboptimal solutions. To alleviate these problems, in this paper we propose a robust and effective model for the MVSC task by jointly incorporating the capped norm, which not only effectively captures the intrinsic global low-rank structure and suppresses the influence of feature outliers by utilizing capped tensor nuclear norm (CTNN), but also integrates the hyper-Laplacian regularization and the spectral embedding through joint optimization to organically benefit from each other. Then, a high computational efficiency algorithm is developed under the alternating direction method of multipliers (ADMM) framework to solve the proposed model. Meanwhile, rigorous mathematical analysis shows that the global optimum of the CTNN regularized least squares subproblem can be analytically determined in closed-form. More importantly, the convergence of the proposed algorithm is theoretically established by proving that the sequence it generates converges to a desirable Karush-Kuhn-Tucker (KKT) point. Finally, extensive experiments on various datasets demonstrate the superiority of the proposed model.

Index Terms—Multi-view subspace clustering, capped tensor nuclear norm, hyper-Laplacian regularization, spectral embedding.

I. INTRODUCTION

CLUSTERING is a fundamental and indispensable technique in data analysis and machine learning [1]. It aims to

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MATLAB code is available at <https://github.com/wangzhi-sw/UFCTNM>.

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partition a set of samples into distinct groups or clusters in an unsupervised manner, relying solely on inherent data structure without leveraging any prior information. In recent years, various clustering algorithms, such as K-means clustering [2], [3] and subspace clustering [4], [5] have been proposed and widely used in image clustering [6], [7], identifying disease [8], [9], person re-identification [10], [11], and outlier detection [12], [13]. Unfortunately, these methods mainly focus on the objects depicted by a single view. In real-world scenarios, data are frequently characterized by diverse types of features or sourced from multiple domains [14], [15], which is referred to as multi-view data. In general, each view provides consensus information and view-specific information, which makes multi-view data inherently more complementary and comprehensive than single-view data. However, these characteristics pose significant challenges in effectively extracting critical information from the original multi-view datasets.

To address this challenge, a straightforward approach is to concatenate all features from different views into a new single-view data, enabling the use of existing advanced single-view clustering algorithms to obtain clustering results for the multi-view data. Although this is a simple approach for multi-view clustering, it often fails to achieve satisfactory performance in most scenarios. This is because the complementary and consistent principle information inherent in the original multi-view data have not been fully taken into account [16]. Most works have demonstrated that such information embedded in multi-view data plays a crucial role in enhancing clustering performance. To alleviate this problem, a large number of multi-view clustering methods [17], [18] have been proposed.

Among them, MVSC methods have achieved state-of-the-art performance by adopting self-representation [19], [20], such as sparse subspace clustering (SSC) [12] and low-rank representation (LRR) [21], which allow them to effectively exploit both the consistency and complementarity across multiple views. The underlying principle of these methods is the assumption that each data point can be represented as a linear combination of other data points. Along this line, most existing MVSC models generally adhere to the similar pipeline [22]: (1) learn view-specific coefficient representations via self-expressiveness property; (2) construct the affinity matrix by merging the learned multi-view representations; (3) perform standard spectral clustering on the affinity matrix to obtain the final clustering result. A core of these steps is how to accurately learn coefficient representations that encode cross-view consistency and view-specific

complementarity, driving recent methodological advancements in MVSC [16], [23], [24]. In [23], Cao et al. leverage the Hilbert Schmidt Independence Criterion (HSIC) [25] as a diversity term to maximize the exploration of complementarity among coefficient representations. In [24], Chen et al. effectively capture the intrinsic low-dimensional structures of multiple views by LRR techniques and diversity regularization, learning complementary and consistent representations simultaneously. In [16], Wang et al. simultaneously impose structural and low-rank constraints on representations utilizing the soft penalty strategy and the non-convex ℓ_q -norm, which comprehensively explore the complementary and consistent information among representations. Although the aforementioned approaches achieve significant performance, they belong to matrix-oriented models [26], failing to explore correlation consistency among views.

Technically, matrix-oriented models exclusively focus on pairwise inter-view correlations through separate processing of individual representation matrices, which ignores high-order correlations among multi-view representations. Due to the fact that the tensor is capable of uncovering high-order information from the multidimensional perspective, several low-rank tensor-oriented MVSC models [27], [28] have emerged to address this limitation by storing all representations into a third-order tensor. Among them, Xie et al. [27] develop tensor singular value decomposition (t-SVD) based MVSC (t-SVD-MSC) model, imposing the tensor nuclear norm (TNN) on a rotated representation tensor to effectively exploit high-order consistency and complementary information. But this model assumes that the features are distributed in multiple linear subspaces, leading to unsatisfactory results when dealing with practical nonlinear data. To solve this problem, hyper-Laplacian regularized multilinear MVSC (HLRM²VS) [28] incorporates geometric constraints to preserve the local geometric structure for nonlinear subspace learning. Nevertheless, these models simply employ TNN to capture the global low-rank structure of the self-representation tensor.

TNN provides a relatively loose approximation to the tensor rank, as it assigns equal penalties to all singular values, which leads to the over-shrinkage of dominant informative components and consequently hampers the discrimination among principal structures, view-specific complementary information, and underlying noise. This fact makes the learned tensor representation inaccurate, resulting in the degradation of cluster performance. To alleviate this problem, numerous non-convex subspace clustering models have been proposed [29], [30], [31]. In [30], Xia et al. apply weighted tensor Schatten- p norm minimization (WTSNM) to MVSC to capture high-order correlation. This approach unifies tensor representation learning and spectral clustering within a single framework to characterize high-order information among views. However, estimating ideal weights remains challenging. Concurrently, Chen et al. [29] propose the partial tubal nuclear norm-regularized multi-view subspace learning (PTN²MSL) model, which minimizes the partial sum of tubal singular values to more effectively capture high-order correlations among views. However, its dependence on fixed truncation parameters limits its adaptability.

Through systematically investigating existing MVSC models, we identify the following limitations still persisting in the learning of coefficient representations:

- 1) While existing MVSC approaches have achieved promising achievements, they have some drawbacks. On one hand, MVSC models based on the convex low-rank approximation fail to fully exploit the structural information embedded in tensor representation under noise interference. On the other hand, some non-convex low-rank regularized MVSC models are computationally expensive, since their non-convex optimization problem has no closed-form solution and rely on numerical algorithms. Thus, an appropriate regularizer should be introduced to capture the intrinsic low-rank structure in the data and ensure exact solutions.
- 2) Hyper-Laplacian regularization [28] preserves intrinsic geometric manifolds across views, while spectral embedding [30] mitigates suboptimal affinity learning through joint representation learning optimization. However, existing models isolate these strengths, which hinder the joint modeling of view-specific local geometric structures and global consistency, limiting clustering performance. To this end, it is necessary to design a unified framework to address this issue.
- 3) The pervasive use of $\ell_{2,1}$ -norm or ℓ_1 -norm for sparse noise modeling imposes uniform sparsity constraints across samples or views, neglecting intrinsic variations in noise distributions. This inflexibility exacerbates the sensitivity to outliers, undermining robustness in practical scenarios. Therefore, it is great significant to introduce a fresh sparsity penalty to achieve robust multi-view learning.

To address aforementioned challenges, this paper proposes a novel MVSC model, named unified framework with capped tensor norm minimization (UFCTNM) for multiview subspace learning. Compared with existing models, UFCTNM has three prominent advantages. First, it effectively captures the intrinsic low-rank structure of multi-view data. Thus, complementary information and high-order correlation among representations can be fully exploited to learn more discriminative representations for MVSC. Second, this model comprehensively considers view-specific local similarity information and effectively avoids suboptimal performance caused by executing the spectral clustering in a separated step, thereby contributing different components to each other and promoting the learned multi-view representations. Third, the proposed model exploits the inherent sparsity in multi-view data through structured error modeling, which accounts for different noise distributions and increases robustness against outliers. Fig. 1 shows the overall framework of the proposed UFCTNM.

In summary, the main contributions of this paper are illustrated as follows.

- 1) A unified model based on capped norm is proposed for the MVSC, which can fully exploit the complementary and consistent connections among multi-view representations, and the robustness of outliers and noise can be enhanced. More importantly, according to our knowledge, the proposed model is the first to integrate the hyper-Laplacian

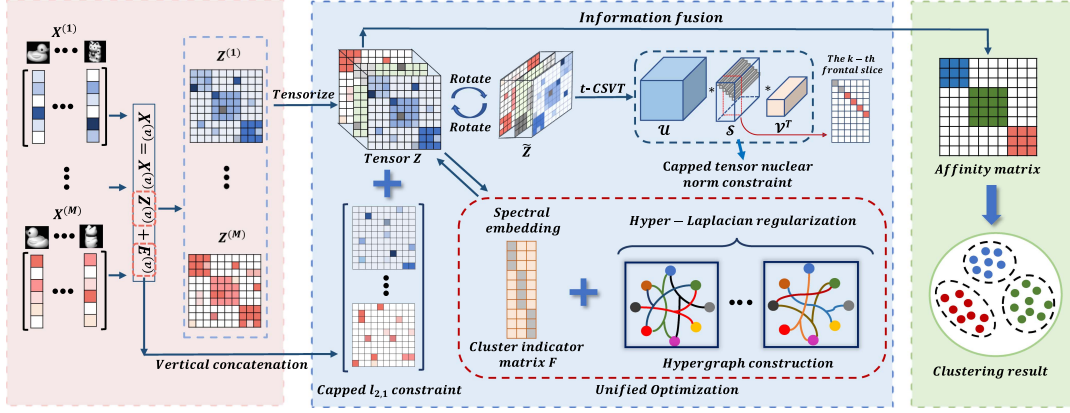


Fig. 1. The overall framework of our proposed UFCTNM. First, the representation matrices $\{Z^{(v)}\}_{v=1}^M$ are explored from the data features and tensor $\mathcal{Z}$ is constructed by stacking all these subspace representations. Then, the low-rank property is depicted by the CTNN, and the hypergraph regularized local geometrical constraint and spectral embedding are performed simultaneously. Moreover, the capped $\ell_{2,1}$ norm is introduced as sparsity penalty to improve model robustness. The ideal tensor representation $\mathcal{Z}$ can be obtained through joint optimization. Finally, the affinity matrix is constructed by merging the learned multi-view representations and the final clustering result can be obtained by performing spectral clustering on the affinity matrix.

regularization and the spectral embedding into one single framework so that they can organically promote each other to serve the clustering task.

- 2) To solve the proposed model, an efficient optimization algorithm is developed based on the ADMM framework. Meanwhile, the capped tensor nuclear norm (CTNN) regularized least squares subproblem is analyzed in detail, showing that the global optimal solution can be directly obtained in a closed-form. This allows our model to be optimized more precisely and efficiently.
- 3) Theoretical convergence guarantee is established, even if the resultant optimization problem is non-convex and multi-block. Specifically, we mathematically prove that the solution sequence generated by the proposed optimization algorithm converges to a Karush-Kuhn-Tucker (KKT) point.
- 4) Extensive experiments on various benchmark datasets comprehensively validate the effectiveness of the proposed model. The results remarkably show the superiority of the proposed model over several related state-of-the-art multi-view clustering models.

The rest of this paper is organized as follows. Section II introduces some related works and preliminaries. In Section III, the UFCTNM model is presented. Then, an efficient optimization algorithm is devised and the convergence of the proposed algorithm is analyzed. In Section IV, the experimental results are presented and analyzed. Finally, some conclusions are derived in Section V.

II. RELATED WORK

In this section, we briefly review the related work, including LRR-based MVSC and manifold regularization. Moreover, some notations and preliminaries used in this paper are summarized here.

A. Multiview Subspace Clustering Based on LRR

Let $X = [x_1, x_2, \dots, x_n] \in \mathbb{R}^{M \times N}$ be a data matrix, where x_i represents the individual feature vector corresponding to the i -th

sample. LRR assumes that data points are drawn from a union of multiple linear subspaces, aiming to recover the underlying subspace structure by considering the following general rank minimization problem

$$\min_Z \text{rank}(Z) + \lambda \|E\|_\ell, \quad \text{s.t.} \quad X = AZ + E, \quad (1)$$

where $Z \in \mathbb{R}^{N \times N}$ is representation matrix, A is a given linear dictionary, $\|E\|_\ell$ represents certain error metric, and $\lambda > 0$ is a balanced parameter. However, such a problem is NP-hard due to the discreteness of the rank function [32]. To address this issue, several convex [33] or non-convex surrogates [34], [35], [36], [37] are employed to approximate the rank function. Despite its effectiveness, LRR remains a single-view method, which cannot fully explore the consistency and complementarity among different views.

To overcome this limitation, LRR-based MVSC models [26], [27], [30], [31] have been developed to exploit the complementary and comprehensive multi-view information. The t-SVD-MSC [27] model is a typical example among them, which imposes TNN on the rotated tensor representation to effectively explore complementary information among views. Given a multi-view dataset $X = [X^{(1)}, X^{(2)}, \dots, X^{(V)}]$, where $X^{(v)} \in \mathbb{R}^{d_v \times N}$ denotes the v -th feature matrix, the t-SVD-MSC model is formulated as

$$\begin{aligned} \min_{Z^{(v)}, E^{(v)}} \quad & \|\mathcal{Z}\|_{\otimes} + \lambda \|E\|_{2,1}, \\ \text{s.t.} \quad & X^{(v)} = X^{(v)} Z^{(v)} + E^{(v)}, v = 1, 2, \dots, V, \\ & \mathcal{Z} = \Phi(Z^{(1)}, Z^{(2)}, \dots, Z^{(V)}), \\ & E = [E^{(1)}; E^{(2)}; \dots; E^{(V)}], \end{aligned} \quad (2)$$

where function $\Phi(\cdot)$ defines the tensor construction strategy, and $E = [E^{(1)}; E^{(2)}; \dots; E^{(V)}]$ indicates the vertically concatenated error matrix. Inspired by t-SVD-MSC, Guo et al. [26] propose a tensor logarithmic Schatten- p norm minimization-based

MVSC (TLSpNM-MSC) model, which can be written as

$$\begin{aligned} \min_{Z^{(v)}, E^{(v)}} \quad & \|Z\|_{L, S_p} + \lambda \sum_{v=1}^V \|E^{(v)}\|_{2,1}, \\ \text{s. t.} \quad & X^{(v)} = X^{(v)} Z^{(v)} + E^{(v)}, v = 1, 2, \dots, V, \\ & Z = \Phi(Z^{(1)}, Z^{(2)}, \dots, Z^{(V)}). \end{aligned} \quad (3)$$

The TLSpNM-MSC model can fully consider the physical difference among singular values and exploit the complementary information across multi-view data. However, these models perform coefficient representation learning and spectral embedding in two separate steps, resulting in suboptimal performance. To solve this problem, Xia et al. [30] propose the WTSNM-MSC model, which is formulated as

$$\begin{aligned} \min_{Z, E^{(v)}, F} \quad & \|Z\|_{w, S_p}^p + \lambda \|E\|_{2,1} + 2\alpha \text{Tr}(F^T L_{\hat{Z}} F), \\ \text{s. t.} \quad & X^{(v)} = X^{(v)} Z^{(v)} + E^{(v)}, v = 1, 2, \dots, V, \\ & Z = \Phi(Z^{(1)}, Z^{(2)}, \dots, Z^{(V)}), \\ & E = [E^{(1)}; E^{(2)}; \dots; E^{(V)}], \end{aligned} \quad (4)$$

where $L_{\hat{Z}} = D_{\hat{Z}} - \hat{Z}$ represents Laplacian matrix, $\hat{Z} = (1/V) \sum_{v=1}^V [(|Z^{(v)}| + |Z^{(v)}|^T)/2]$ and $D_{\hat{Z}}$ is a diagonal matrix, whose diagonal elements are $D_{\hat{Z}}(i, i) = \sum_j (\hat{Z}(i, j) + \hat{Z}(j, i))$. F denotes cluster indicator matrix. By integrating representation learning and spectral clustering into a unified framework, the final fusion similarity matrix resulting from this model can effectively characterize the cluster structure. In summary, existing LRR-based MVSC models show significant advantages in integrating essential multi-view information, particularly in preserving structural consistency and exploring complementary information, which deserves further study.

B. Manifold Regularization

The majority of LRR-based MVSC models are primarily restricted to handling features that are distributed across multiple linear subspaces [27], [38]. However, their linearity assumption limits their applicability to real-world data with complex nonlinear manifold structures, as they fail to preserve local geometric relationships and global nonlinear dependencies. To alleviate this problem, in recent years, a large number of manifold learning models have been proposed to preserve the local geometric structure of data embedded in high-dimensional space, including local linear embedding [39], locality preserving projection [40], and manifold regularization [41]. Among them, manifold regularization models are more convenient to impose local geometric constraints on the feature space compared with other models [28].

Numerous studies have introduced Laplacian regularizers to preserve the local geometric structure of multiple features [28], [31], [42]. In [42], Lei et al. design a manifold regularized MVSC model to more effectively integrate relevant and complementary information from different perspectives. However, the data structure is actually complex and this model may not be able

to fully utilize the view-specific local similarity information under the constraint of Laplace regularization. To overcome this limitation, Xie et al. [28] propose to apply hyper-Laplacian regularization to the self-representation matrix of a specific view, which jointly learns the correlation and local geometric structure between views. Similarly, Wang et al. [31] propose a hyper-Laplacian regularized non-convex low-rank representation (HNLR) model to capture the local high-order geometrical structure and the global low-rank structure of views simultaneously. Despite these models apply the global and local relevant information of views based on LRR and manifold constraint, the structure learning of the representations still needs to be further improved.

C. Preliminaries

For a third-order tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, $\bar{\mathcal{X}}$ represents discrete fast Fourier transform (FFT) of $\mathcal{X}$ along the third dimension, i.e., $\bar{\mathcal{X}} = \text{fft}(\mathcal{X}, [], 3)$ and $\mathcal{X} = \text{ifft}(\bar{\mathcal{X}}, [], 3)$. For tensor $\mathcal{X}$, its **transpose** is $\mathcal{X}^T \in \mathbb{R}^{n_2 \times n_1 \times n_3}$, which obtained by transposing each frontal slice and then reversing the order of the transposed frontal slices 2 to n_3 . A tensor is a f -diagonal tensor if and only if its frontal slices are all diagonal matrices. The tensor $\mathcal{I} \in \mathbb{R}^{n_1 \times n_1 \times n_3}$ is an **identity tensor** if its first frontal slice is an identity matrix and other frontal slices are zero.

Now, given tensors $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and $\mathcal{Y} \in \mathbb{R}^{n_2 \times n_4 \times n_3}$, we introduce following related operations. The block circular matrix $\text{bcirc}(\mathcal{X})$, block diagonal matrix $\text{bdiag}(\mathcal{X})$ and block vectorization $\text{bvec}(\mathcal{X})$ can be constructed by

$$\text{bcirc}(\mathcal{X}) = \begin{bmatrix} \mathcal{X}^{(1)} & \mathcal{X}^{(n_3)} & \dots & \mathcal{X}^{(2)} \\ \mathcal{X}^{(2)} & \mathcal{X}^{(1)} & \dots & \mathcal{X}^{(3)} \\ \vdots & \vdots & \ddots & \vdots \\ \mathcal{X}^{(n_3)} & \mathcal{X}^{(n_3-1)} & \dots & \mathcal{X}^{(1)} \end{bmatrix}, \quad (5)$$

$$\text{bdiag}(\mathcal{X}) = \begin{bmatrix} \mathcal{X}^{(1)} & & & \\ & \mathcal{X}^{(2)} & & \\ & & \ddots & \\ & & & \mathcal{X}^{(n_3)} \end{bmatrix}, \quad (6)$$

$$\text{bvec}(\mathcal{X}) = [\mathcal{X}^{(1)}; \dots; \mathcal{X}^{(n_3)}]. \quad (7)$$

The inverse operations of bdiag and bvec are defined as $\text{bdfold}(\text{bdiag}(\mathcal{X})) = \mathcal{X}$ and $\text{bvfold}(\text{bvec}(\mathcal{X})) = \mathcal{X}$, respectively. The **t-product** between $\mathcal{X}$ and $\mathcal{Y}$ is $\mathcal{Z} \in \mathbb{R}^{n_1 \times n_4 \times n_3}$, that is

$$\mathcal{Z} = \mathcal{X} * \mathcal{Y} = \text{bvfold}\{\text{bcirc}(\mathcal{X})\text{bvec}(\mathcal{Y})\}. \quad (8)$$

Tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_1 \times n_3}$ is **orthogonal** if it satisfies $\mathcal{X}^T * \mathcal{X} = \mathcal{X} * \mathcal{X}^T = \mathcal{I}$. Based on these operations, we have some definitions as follows.

Definition 1 (t-SVD [43]). Given tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, its tensor Singular Value Decomposition (t-SVD) is defined as

$$\mathcal{X} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T, \quad (9)$$

where $\mathcal{U} \in \mathbb{R}^{n_1 \times n_1 \times n_3}$ and $\mathcal{V} \in \mathbb{R}^{n_2 \times n_2 \times n_3}$ are orthogonal tensors, $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is an f -diagonal tensor.

TABLE I
 THE SUMMARIZATION OF NOTATIONS

Notation	Description
$\mathcal{X}, X, \mathbf{x}, x$	tensor, matrix, vector, scalar
v, V	The v -th view and the number of views
d_v, n	The dimension of v -th view and the number of sample
$X(i, j)$	The (i, j) -th element of X
$\mathcal{X}(i, j, k)$	The (i, j, k) -th element of $\mathcal{X}$
X^T	The transpose of X
$\sigma_i(X)$	The i -th leading singular value of X
$tr(X)$	The trace of X
$sign(x)$	The sign function about x
$\mathcal{X}^{(k)}$	The k -th frontal slice of the tensor $\mathcal{X}$
$X^{(v)}$	The feature matrix of the v -th view
$\ \mathbf{x}\ _2$	The ℓ_2 -norm of vector $\mathbf{x}$
$\ X\ _F = \sqrt{tr(X^T X)}$	Frobenius norm of X
$\ X\ _{2,1} = \sum_j \ X(:, j)\ _2$	Frobenius norm, $\ell_{2,1}$ -norm
$\ X\ _\infty = \max_{ij} X(i, j) $	Infinity norm of X

Definition 2 (TNN [44]). Let $\mathcal{X} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$ be the t-SVD of $\mathcal{X}$, then the tensor nuclear norm of $\mathcal{X}$ can be formulated as follows

$$\|\mathcal{X}\|_{\otimes} = \frac{1}{n_3} \sum_{i=1}^{\min(n_1, n_2)} \sum_{k=1}^{n_3} |\bar{S}^{(k)}(i, i, k)|. \quad (10)$$

Finally, we give some preliminary knowledge of hypergraphs. Given a hypergraph $G = (V, E, W)$, where V represents a set of vertices, E is a hyperedge set e of V , the V and E satisfy the relationship $U_{e \in E} = V$, and the positive weight $w(e)$ is the element of matrix W . An incidence matrix $H \in \mathbb{R}^{|V| \times |E|}$ in the hypergraph G denotes the relationship between the vertices and the hyperedges, the elements of which are defined as

$$h(\mathbf{v}_i, \mathbf{e}_j) = \begin{cases} 1, & \text{if } \mathbf{v}_i \in \mathbf{e}_j, \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

Let $d(\mathbf{e}_j) = \sum_{\mathbf{v}_i \in V} h(\mathbf{v}_i, \mathbf{e}_j)$ be the degree of hyperedge $\mathbf{e}_j \in E$ and D_E be the diagonal matrix whose diagonal elements correspond to the degree of each hyperedge. Similarly, let $d(\mathbf{v}_i) = \sum_{\mathbf{e}_j \in E} w(\mathbf{e}_j) h(\mathbf{v}_i, \mathbf{e}_j)$ be the degree of vertex $\mathbf{v}_i \in V$ and D_V be the diagonal matrix whose diagonal elements correspond to the degree of each vertex. Then the unnormalized hyper-Laplacian matrix of hypergraph can be defined as

$$L_h = D_V - H W D_E^{-1} H^T. \quad (12)$$

The notations used in this work are summarized in Table I.

III. THE PROPOSED MODEL

In this section, we first formulate the proposed model by jointly incorporating the capped norm while unifying the representation learning, hyper-Laplacian regularization, and spectral embedding into a single framework. Then, the solution of CTNN is analyzed, and an efficient algorithm with the convergence guarantee is developed to solve the resultant optimization problem.

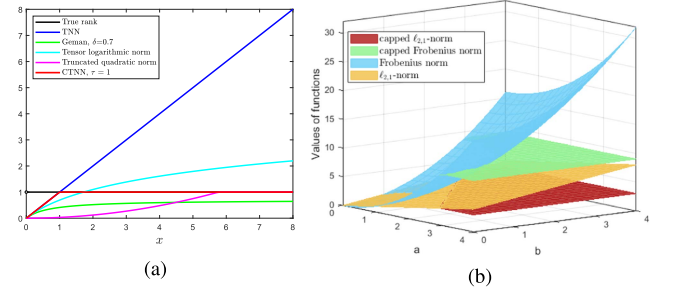


Fig. 2. Illustration of rank approximate and outliers resistance functions. (a) Comparison of different methods to approximate the true rank function on the scalar case. (b) Comparison of different functions for outliers resistance on vector $[a, b]$, where $0 \leq a \leq 4$ and $0 \leq b \leq 4$.

A. Problem Formulation

The majority of existing MVSC models employ convex or non-convex surrogate for rank approximation in self-representation tensor learning, inducing biased rank estimation or computational inefficiency. Moreover, during sparse error modeling, existing models exhibit high susceptibility to extreme outliers, which significantly compromises their outlier suppression capability. To overcome these limitations, in this work we propose a robust and effective model for the MVSC task by jointly incorporating the capped norm, which can be formulated as follows

$$\begin{aligned} \min_{Z^{(v)}, E^{(v)}} \quad & \|Z\|_{C_*^\tau} + \lambda \|E\|_{C_{2,1}^\theta}, \\ \text{s. t.} \quad & X^{(v)} = X^{(v)} Z^{(v)} + E^{(v)}, v = 1, 2, \dots, V, \\ & Z = \Phi(Z^{(1)}, Z^{(2)}, \dots, Z^{(V)}), \\ & E = [E^{(1)}; E^{(2)}; \dots; E^{(V)}]. \end{aligned} \quad (13)$$

The first term in the proposed model is the so-called CTNN, which is mathematically defined as

$$\|Z\|_{C_*^\tau} = \frac{1}{n_3} \sum_{j=1}^{n_3} \sum_{i=1}^{\min(n_1, n_2)} \min(\sigma_i(\bar{Z}^{(j)}), \tau), \quad (14)$$

where $\tau \geq 0$ is a threshold. The second term is the so-called capped $\ell_{2,1}$ -norm [45], which is defined as

$$\|E\|_{C_{2,1}^\theta} = \sum_{j=1}^n \min\{\|E(:, j)\|_2, \theta\}, \quad (15)$$

where $\theta \geq 0$ is a parameter.

Compared with the existing MVSC models, the proposed model simultaneously achieves the following three advantages.

- 1) *It fully explore complementarity and consistency among multiple views:* Since our model does not minimize the variance of data distribution within the target rank, which suppress the excessively larger singular values and punish the insignificant ones, as shown in Fig. 2(a). This advantage helps to make sure that the learned tensor representation has a spatial low-rank structure, then the complementary and consistent information based on global perspective can be explored across all views. Thus,

it can effectively ensure the high-order consistency and fully capture the complementarity among multi-views.

- 2) *It can be solved by an efficient algorithm:* Since we showed that the proximal operator of CTNN allows a closed-form solution. In previous works [46], [47], such operator has to be solved by inefficient iterative algorithms, such as reweighted method [47] and DC programming [48], which limits the application of CTNN-based models. To this end, we proved that the global optimal solution can be obtained directly in a single step. Thus, the algorithm we designed is efficient and fast for solving the proposed model.
- 3) *It can effectively suppress outliers and noise:* Since the value of the error metric introduced by the proposed model does not grow linearly with increasing error and the value of its column norm will be constant when it is larger than the threshold, the influence of extreme outliers on clustering is limited within a certain range, as illustrated in Fig. 2(b).

In addition, it can also suppress the influence of feature noise, which is the same as traditional measurements. Hence, the proposed model can ensure robustness to both extreme outliers (excluding randomly distributed outliers) and noise across different views.

The preservation of the local geometric structure means that the model retains local adjacency relationships among multiple samples to effectively capture nonlinear manifold characteristic and high-order local correlation of them, while the combination of spectral embedding and representation learning effectively avoids the suboptimality caused by executing these steps separately. However, existing models fail to unify these advantages, which hinders the joint optimization of preserving view-specific local similarity information and encoding discriminative information in representations. To address this issue, we propose the following unified framework for MVSC.

$$\begin{aligned}
\min_{Z^{(v)}, E^{(v)}} \quad & \|Z\|_{C_{\tau}} + \lambda_1 \|E\|_{C_{2,1}^g} + 2\alpha \text{tr}(F^T L_Z F) \\
& + \lambda_2 \sum_{v=1}^V \text{tr} \left(Z^{(v)} L_h^{(v)} Z^{(v)T} \right), \\
\text{s. t.} \quad & X^{(v)} = X^{(v)} Z^{(v)} + E^{(v)}, v = 1, 2, \dots, V, \\
& Z = \Phi(Z^{(1)}, Z^{(2)}, \dots, Z^{(V)}), \\
& E = [E^{(1)}; E^{(2)}; \dots; E^{(V)}], \quad (16)
\end{aligned}$$

where λ_1 , λ_2 and α are balance parameters, $L_h^{(v)}$ is the view-specific hyper-Laplacian matrix with respect to the self-representation matrix $Z^{(v)}$. $F \in \mathbb{R}^{n \times K}$ represents the cluster indicator matrix, where K is the number of clusters. To the best of our knowledge, the proposed model is the first attempt to unify the representation learning, hyper-Laplacian regularization and spectral embedding into a single framework for MVSC, which fully explore all-round relation among data points and utilizes the dependence tensor representation and cluster indicator matrix. In summary, a robust tensor representation that preserves the intrinsic non-linear geometric information while decoding the discriminative information can be learned in proposed model,

which well explore consistent and complementary connections among multiple views. Thus, the affinity matrix constructed from the tensor $\mathcal{Z}$ can characterize clustering-friendly structure [49], ultimately improving the clustering performance.

B. Optimization Strategy

Due to the non-convexity of the proposed model, it is intractable to solve it directly by the off-the-shelf tools. To address this problem, an effective and efficient algorithm is proposed using the framework of ADMM. We introduce two auxiliary variables $\mathcal{G}$ and $Q^{(v)}$ for the separability of $\mathcal{Z}$ and $Z^{(v)}$, respectively. Then the proposed model can be transferred to the following unconstrained one.

$$\begin{aligned}
\mathcal{L}(Z^{(v)}, E^{(v)}, Q^{(v)}, \mathcal{G}, F; \mathcal{W}, B^{(v)}, Y^{(v)}, \rho, \mu_1, \mu_2) \\
= \|\mathcal{G}\|_{C_{\tau}} + \lambda_1 \|E\|_{C_{2,1}^g} + \sum_{v=1}^V \lambda_2 \text{tr} \left(Q^{(v)} L_h^{(v)} Q^{(v)T} \right) \\
+ 2\alpha \text{tr}(F^T L_Z F) + \langle \mathcal{W}, Z - \mathcal{G} \rangle + \frac{\rho}{2} \|Z - \mathcal{G}\|_F^2 \\
+ \sum_{v=1}^V \left(\langle B^{(v)}, Q^{(v)} - Z^{(v)} \rangle + \frac{\mu_2}{2} \|Q^{(v)} - Z^{(v)}\|_F^2 \right. \\
+ \frac{\mu_1}{2} \|X^{(v)} - X^{(v)} Z^{(v)} - E^{(v)}\|_F^2 \\
\left. + \langle Y^{(v)}, X^{(v)} - X^{(v)} Z^{(v)} - E^{(v)} \rangle \right), \quad (17)
\end{aligned}$$

where $\mathcal{W}$, $B^{(v)}$ and $Y^{(v)}$ represent Lagrange multipliers, μ_1 , μ_2 and ρ denotes the penalty parameters. Then, we update each variable alternately by fixing other.

$Z^{(v)}$ -subproblem: Fixing the other variables, according to $\Phi_{(v)}^{-1}(\mathcal{W}) = W^{(v)}$ and $\Phi_{(v)}^{-1}(\mathcal{G}) = G^{(v)}$, the subproblem involving with $Z^{(v)}$ can be written as

$$\begin{aligned}
Z^{(v)} = \arg \min_{Z^{(v)}} \quad & 2\alpha \text{tr}(F^T L_Z F) \\
& + \langle W^{(v)}, Z^{(v)} - G^{(v)} \rangle + \frac{\rho}{2} \|Z^{(v)} - G^{(v)}\|_F^2 \\
& + \langle B^{(v)}, Q^{(v)} - Z^{(v)} \rangle + \frac{\mu_2}{2} \|Q^{(v)} - Z^{(v)}\|_F^2 \\
& + \frac{\mu_1}{2} \|X^{(v)} - X^{(v)} Z^{(v)} - E^{(v)}\|_F^2 \\
& + \langle Y^{(v)}, X^{(v)} - X^{(v)} Z^{(v)} - E^{(v)} \rangle. \quad (18)
\end{aligned}$$

Obviously, (18) is smooth and convex. Denote by $P = [P_1, \dots, P_j, \dots, P_N]$, where $P_j = [\|F(1, :) - F(j, :)\|_2^2; \dots; \|F(N, :) - F(j, :)\|_2^2]$ and $F(j, :)$ is the j -th row of F . Then, we have

$$\begin{aligned}
2\text{tr}(F^T L_Z F) &= \text{tr}(P^T \hat{Z}) \\
&= \text{tr} \left(P^T \left(\frac{1}{V} \sum_{v=1}^V \frac{|Z^{(v)}| + |Z^{(v)}|^T}{2} \right) \right) \\
&= \frac{1}{2V} \text{tr} \left(P^T |Z^{(v)}| + P^T |Z^{(v)}|^T \right). \quad (19)
\end{aligned}$$

(18) can be solved directly by substituting (19) into (18) and its closed-form solution can be obtained as

$$\begin{aligned} Z^{(v)} = & \left(\mu_2 I + \rho I + \mu_1 X^{(v)T} X^{(v)} \right)^{-1} \\ & \left(\mu_1 X^{(v)T} X^{(v)} + X^{(v)T} Y^{(v)} + B^{(v)} + \mu_2 Q^{(v)} \right. \\ & \left. + \rho G^{(v)} - \mu_1 X^{(v)T} E^{(v)} - W^{(v)} \right. \\ & \left. - \frac{\alpha}{2V} \left(P \odot \text{sign}(Z^{(v)}) + P^T \odot \text{sign}(Z^{(v)T}) \right) \right), \end{aligned} \quad (20)$$

where $\text{sign}(\cdot)$ is a sign function.

E-subproblem: Fixing the other variables, the subproblem involving with E can be written as

$$\arg \min_{E^{(v)}} \frac{\lambda_1}{\mu_1} \|E\|_{C_{2,1}^\theta} + \frac{1}{2} \|E - D\|_F^2, \quad (21)$$

where D concatenation vertically along the column of $D^{(v)}$, i.e., $D = [D^{(1)}; D^{(2)}; \dots; D^{(V)}]$ and $D^{(v)} = X^{(v)} - X^{(v)} Z^{(v)} + \mu_1^{-1} Y^{(v)}$. The j -th column of E can be obtained by [50]

$$E(:, j) = \begin{cases} \frac{x_1 D(:, j)}{\|D(:, j)\|_2}, & \text{if } h_j(x_1) \leq h_j(x_2), \\ \frac{x_2 D(:, j)}{\|D(:, j)\|_2}, & \text{otherwise,} \end{cases} \quad (22)$$

where $x_1 = \max(\theta, \|D(:, j)\|_2)$, $x_2 = \min(\theta, [\|D(:, j)\|_2 - \frac{\lambda_1}{\mu_1}]_+)$ for the positive value $\|D(:, j)\|_2$ and $h_j(x) = 0.5(x - \|D(:, j)\|_2)^2 + \lambda_1/\mu_1 \min(|x|, \theta)$.

Q^(v)-subproblem: Fixing the other variables, the subproblem involving with $Q^{(v)}$ can be written as

$$\begin{aligned} Q^{(v)} = & \arg \min_{Q^{(v)}} \lambda_2 \text{tr} \left(Q^{(v)} L_h^{(v)} Q^{(v)T} \right) \\ & + \left\langle B^{(v)}, Q^{(v)} - Z^{(v)} \right\rangle + \frac{\mu_2}{2} \|Q^{(v)} - Z^{(v)}\|_F^2. \end{aligned} \quad (23)$$

The convex quadratic (23) can be easily solved by setting the derivative with respect to $Q^{(v)}$ to zero, we get

$$Q^{(v)} = (\mu_2 Z^{(v)} - B^{(v)}) (2\lambda_2 L_h^{(v)} + \mu_2 I)^{-1}. \quad (24)$$

G-subproblem: Fixing the other variables, the subproblem involving with G can be written as

$$\begin{aligned} G = & \arg \min_G \|G\|_{C_*^\tau} + \langle W, Z - G \rangle + \frac{\rho}{2} \|Z - G\|_F^2 \\ = & \arg \min_G \|G\|_{C_*^\tau} + \frac{\rho}{2} \|G - \left(Z + \frac{1}{\rho} W \right)\|_F^2. \end{aligned} \quad (25)$$

However, it is challenging to directly obtain the global optimal solution of the CTNN regularized least squares subproblem (25). To this end, we propose the following theorem¹ to address this issue.

Theorem 1. Suppose $\mathcal{Y} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ with t-SVD $\mathcal{Y} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$ and $\xi > 0$. Then, the optimal solution to

$$\arg \min_{\mathcal{X}} \frac{1}{2} \|\mathcal{X} - \mathcal{Y}\|_F^2 + \xi \|\mathcal{X}\|_{C_*^\tau} \quad (26)$$

is given by

$$\mathcal{X}^* = \mathcal{U} * \text{ifft}(\mathcal{D}_{\xi, \tau}(\bar{\mathcal{S}}), [], 3) * \mathcal{V}^T, \quad (27)$$

where $\text{ifft}(\mathcal{D}_{\xi, \tau}(\bar{\mathcal{S}}), [], 3) \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is an f -diagonal tensor, and

$$\mathcal{D}_{\xi, \tau}(\bar{\mathcal{S}}_{ii}^{(j)}) = \begin{cases} x_1, & \text{if } f_i(x_1) \leq f_i(x_2) \\ x_2, & \text{otherwise} \end{cases} \quad (28)$$

with $x_1 = \max(\tau, \bar{\mathcal{S}}_{ii}^{(j)})$, $x_2 = \min(\tau, [\bar{\mathcal{S}}_{ii}^{(j)} - \xi]_+)$, and $f_i(x) = 0.5(x - \bar{\mathcal{S}}_{ii}^{(j)})^2 + \xi \min(|x|, \tau)$.

Let $\mathcal{F} = \mathcal{Z} + \rho^{-1} \mathcal{W}$ and the t-SVD of $\mathcal{F}$ be $\mathcal{U}_F * \mathcal{S}_F * \mathcal{V}_F^T$. According to Theorem 1, the global optimum of subproblem (25) can be achieved by

$$G = \mathcal{U}_F * \text{ifft}(\mathcal{D}_{\frac{\rho}{2}, \tau}(\bar{\mathcal{S}}_F), [], 3) * \mathcal{V}_F^T. \quad (29)$$

F-subproblem: By fixing other variables, the subproblem involving with F can be written as

$$\begin{aligned} F = & \arg \min_F \text{tr}(F^T L_{\hat{Z}} F), \\ \text{s. t. } & F^T F = I, F \in \mathbb{R}^{n \times K}, \end{aligned} \quad (30)$$

where the solution F is obtained by computing the eigenvectors corresponding to the smallest K eigenvalues of the Laplacian matrix $L_{\hat{Z}}$ [51].

In addition, the multipliers $Y^{(v)}$, $B^{(v)}$, $\mathcal{W}$ and the penalty parameter μ_1 , μ_2 , ρ can be update by

$$\begin{cases} Y^{(v)} = Y^{(v)} + \mu_1 (X^{(v)} - X^{(v)} Z^{(v)} - E^{(v)}), & (31) \\ B^{(v)} = B^{(v)} + \mu_2 (Q^{(v)} - Z^{(v)}), & (32) \\ \mathcal{W} = \mathcal{W} + \rho (\mathcal{Z} - G), & (33) \\ \mu_i = \min(\eta \mu_i, \mu_{\max}), \quad i = 1, 2 & (34) \\ \rho = \min(\eta \rho, \rho_{\max}), & (35) \end{cases}$$

where $\eta > 1$ is the step size, $\mu_{\max}$ and $\rho_{\max}$ give the upper bound for μ and ρ , respectively. Finally, the complete algorithm for solving the proposed model is summarized in Algorithm 1.

C. Computational Complexity Analysis

The computational complexity of Algorithm 1 mainly focuses on the construction of hyper-Laplacian $L_h^{(v)}$ and the update of variables $Z^{(v)}$, E , $Q^{(v)}$, G and F . In step 2, the hyper-Laplacian takes $\mathcal{O}(nV^2 \log(n))$ for all the views. In step 3, updating $Z^{(v)}$ takes $\mathcal{O}(n^3 + n^2 d_v)$ time because it requires the matrix inverse operation. In step 4, the complexity generated by updating $E \in \mathbb{R}^{d_v V \times n}$ is $\mathcal{O}(d_v nV)$. The complexity of step 5 is $\mathcal{O}(k \log^c(k))$ for a constant c [52] when the Laplacian matrix has k nonzero entries. For tensor $G \in \mathbb{R}^{n \times V \times n}$, its updating relies on operations of SVD, FFT and inverse FFT, thus the complexity is $\mathcal{O}(n^2 V \log(n) + n^2 \sim V^2)$. In addition, the complexity of updating F is $\mathcal{O}(n^3)$. Because $V \ll d_v, n$, the overall computational complexity of Algorithm 1 is $\mathcal{O}(T(n^3 + n^2 V \log(n) + n^2 d_v))$ after T iterations.

D. Convergence Analysis

The ADMM algorithm has been extensively studied for solving convex problems, and its convergence properties have been

¹ All the proofs can be found in the Supplementary Materials.

Algorithm 1: The Algorithm for Solving Problem (16).

Input: Multi-view data $\{X^{(1)}, X^{(2)}, \dots, X^{(V)}\}$, parameter $\lambda_1, \lambda_2, \theta, \alpha, \tau$ and cluster number K .
Initialized: $Z^{(v)} = Q^{(v)} = B^{(v)} = \mathbf{0}$, $E^{(v)} = Y^{(v)} = \mathbf{0}$, $\mathcal{G} = \mathcal{W} = \mathbf{0}$, $\forall v \in \{1, \dots, V\}$; $\mu_1 = \mu_2 = \rho = 10^{-5}$, $\eta = 2$, $\mu_{\max} = \rho_{\max} = 10^{12}$; $\varepsilon = 10^{-7}$.
1: **while** not converge **do**
2: Construct hyper-Laplacian matrices $\{L_h^{(v)}\}_{v=1}^V$ by using k -nearest from $\{Z^{(v)}\}_{v=1}^V$;
3: Update $\{Z^{(v)}\}_{v=1}^V$ by (20);
4: Update E by (22)
5: Update $\{Q^{(v)}\}_{v=1}^V$ by (23);
6: Update $\mathcal{G}$ by (29);
7: Update F by (30);
8: Update $\{Y^{(v)}\}_{v=1}^V$, $\{B^{(v)}\}_{v=1}^V$, $\mathcal{W}$, μ_1 , μ_2 and ρ by (31) - (35);
9: Obtain $\mathcal{Z} = \Phi(Z^{(1)}, Z^{(2)}, \dots, Z^{(V)})$;
10: Obtain $(G^{(1)}, G^{(2)}, \dots, G^{(V)}) = \Phi^{-1}(\mathcal{G})$;
11: Check the convergence conditions:
 $\|X^{(v)} - X^{(v)}Z^{(v)} - E^{(v)}\|_{\infty} < \varepsilon$, $\|Q^{(v)} - Z^{(v)}\|_{\infty} < \varepsilon$,
 $\|Z^{(v)} - G^{(v)}\|_{\infty} < \varepsilon$;
12: **end while**
13: Obtain the affinity matrix by
 $S = \frac{1}{V} \sum_{v=1}^V (|Z^{(v)}| + |Z^{(v)T}|)$;
Output: The clustering result C by performing spectral clustering on S .

well established. Recent research [53] has generalized the convergence analysis of two-block ADMM to non-convex optimization problems. However, there are significant challenges in proving the convergence of ADMM-based algorithms for multi-block problems. Thus, it is hard to theoretically justify the convergence of our proposed algorithm. Here, we still provide a rigorous theoretical guarantee for the proposed algorithm as shown in Theorem 2, which demonstrates that any accumulation point generated by the proposed algorithm satisfies the first-order KKT conditions.

Theorem 2. Let the sequence $\{\Gamma_k = (Z_k^{(v)}, E_k^{(v)}, Q_k^{(v)}, \mathcal{G}_k; B_k^{(v)}, Y_k^{(v)}, \mathcal{W}_k)\}_{k=1}^{\infty}$ be generated by Algorithm 1. If $\lim_{k \rightarrow \infty} \{t_k \circ (\mathcal{N}_{k+1}^{(v)} - \mathcal{N}_k^{(v)})\} = \mathbf{0}$ with $\mathcal{N}_k^{(v)} = (E_k^{(v)}, Q_k^{(v)}, G_k^{(v)})$ and $t_k = (\mu_{1,k}, \mu_{2,k}, \rho_k)$ holds, the sequence $\{\Gamma_k\}$ satisfies the following three properties:

- 1) The sequence $\{\Gamma_k\}_{k=1}^{\infty}$ is bounded.
- 2) Any accumulation point Γ_* of $\{\Gamma_k\}_{k=1}^{\infty}$ satisfies first-order optimality KKT conditions. Specially, $\{\Gamma_k\}_{k=1}^{\infty}$ converges to a KKT point, i.e.,

$$\begin{cases} \mathbf{0} = X^{(v)} - X^{(v)}Z_*^{(v)} - E_*^{(v)}, \\ \mathbf{0} = \mathcal{Z}_* - \mathcal{G}_*, \\ \mathbf{0} = Q_*^{(v)} - Z_*^{(v)}, \\ \mathbf{0} \in \lambda_1 \partial \|E_*^{(v)}\|_{C_{2,1}^{\theta}}, \\ \mathbf{0} \in \lambda_2 \partial \|\mathcal{G}_*\|_{C_{\tau}^{\theta}} - \mathcal{W}_*, \\ \mathbf{0} = B_*^{(v)} - 2\lambda_2 Q_*^{(v)} L_{h,*}^{(v)}, \\ \mathbf{0} = X^{(v)T} Y_*^{(v)} + B_*^{(v)} - \mathcal{W}_* - \Delta. \end{cases} \quad (36)$$

where $\Delta = \frac{\alpha}{2^m} (P \odot \text{sign}(Z_*^{(v)}) + P^T \odot \text{sign}(Z_*^{(v)T}))$.

- 3) $\{Z_t^{(v)}\}$, $\{E_t^{(v)}\}$, $\{Q_t^{(v)}\}$, $\{\mathcal{G}_t\}$ are Cauchy sequences, and thus converge to their critical points.

From Theorem 2, we can see that the variable sequences generated by the proposed algorithm are bounded and they converge to a stationary KKT critical point. In addition, we have rigorously established the convergence of the objective function, which is detailed in the supplementary material.

This established convergence result confirms the effectiveness of the proposed algorithm in solving the optimization problem arising from the proposed model. Furthermore, the above KKT conditions can help us determine the stopping conditions for Algorithm 1, which are $\|X^{(v)} - X^{(v)}Z^{(v)} - E^{(v)}\|_{\infty} < \varepsilon$, $\|Q^{(v)} - Z^{(v)}\|_{\infty} < \varepsilon$, $\|Z^{(v)} - G^{(v)}\|_{\infty} < \varepsilon$.

E. Differences From Several Existing Works

The proposed model fundamentally advances the previous LRR-based MVSC models, including t-SVD-MSC [27], HLR-M² VS [28], PTN² MSL [29], WTSNM [30], in three critical aspects. First, unlike t-SVD-MSC and HLR-M² VS, which utilize TNN as low-rank regularizer and fail to fully exploit the structural information, our model introduces CTNN to capture the intrinsic low-rank structure. At the same time, we also observe that the number of truncation in CTNN varies with iterations, while the PTNN in PTN² MSL has dependency on a fixed truncation parameter, which means that CTNN is more flexible than PTNN. Second, while existing methods either neglect view-specific local geometry or decouple representation learning from spectral embedding, the proposed model designs a unified framework to jointly preserve intra-view geometric structures and encode discriminative information in representations. Third, the proposed model introduces a fresh sparsity penalty to limit the influence of extreme outliers on clustering. In contrast, the t-SVD-MSC, HLR-M² VS, PTN² MSL and WTSNM models all use $\ell_{2,1}$ -norm for sparse modeling, which conventionally undermines outlier suppression capability due to the susceptibility to extreme-norm outliers. In summary, the proposed model has a significant difference from other related methods.

IV. EXPERIMENTS

In this section, we conduct experiments on nine benchmark datasets and compare with several state-of-the-art baselines to verify the effectiveness and robustness of the model we proposed.

A. Experimental Settings

1) **Datasets:** To verify the effectiveness of the proposed method, nine challenging real-world datasets are adapted, including six types, i.e., text, face, number, scene, object and plant. The statistics of these datasets are shown in Table II. Moreover, the brief descriptions of these datasets are summarized as follows.

- **20Newsgroups** is a subset of original 20 Newsgroup dataset. It contains 500 newsgroup documents with three views and each document carries five class labels.

TABLE II
STATISTICAL INFORMATION FOR THE BENCHMARK DATABASES

Type	Dataset	Sample	Clusters	View
Text	20NewsGroups	500	5	3
Face	Yale	165	15	3
	ORL	400	40	3
	Notting-Hill	4660	5	3
Digit	UCI-Digits	2000	10	3
Scene	MITIndoor	5360	67	4
Object	ALOI	1080	10	4
	COIL20	1440	20	3
Plant	100leaves	1660	100	3

- *Yale* contains 165 Gy scale face images from 15 different people, each with 11 samples from different facial expressions or configurations. Three types of features (Intensity, LBP and Gabor) are extracted in this dataset.
- *ORL* consists of 400 face images from 40 individuals, with each person captured under 10 different illumination and facial motions. Intensity, LBP, and Gabor features are constructed in this dataset.
- *Notting-Hill* is derived from video clips from the movie of the same name. This dataset contains five main casts face images, including 4460 faces from 76 tracks. Three features (Intensity, LBP and Gabor) are extracted.
- *UCI-Digits* contains images of handwritten digits from 0 to 9, with 200 images for each digit and 2000 images in total. It provides three views including 76 Fourier coefficients with the character shapes, 216 profile correlations and 64 Karhunen-Love coefficients.
- *MITIndoor* is an indoor image dataset consisting of 5360 images with 67 different categories. Besides PHOW, CENTRIST and LBP features, the deep feature VGG-VD pre-trained on ILSVRC12 [54] is also extracted.
- *ALOI* contains 110,250 color images of 1,000 small objects captured under various rotation angles and illumination conditions. We select the first 10 classes with RGB color histograms, color similarity, HSB color histograms, and Haralick features.
- *COIL20* consists of 1440 images with 20 classes captured from various angles. Each category includes 72 images and three features (Intensity, LBP, and Gabor) are extracted for clustering.
- *100leaves* is a leaf recognition dataset for botany applications. It contains 100 different plant leaf categories and 1,600 images, with at least 16 images for each category. The texture, shape and margin of each leaf are extracted as three different views.

2) *Competitors*: To verify the superiority of our proposed model, fourteen representative multi-view clustering models are selected as competitors. Four of them are the matrix-oriented models, including DiMSC [23], MLRR [24], LSGMC [55] and FAMvC [56]. The rest are tensor-based models, including LT-MSC [38], ETLMSC [57], HLR-M² VS [28], GNLTA [22], WTSNM [30], t-SVD-MSC [27], EMKIC [58], L1SL- $\ell_{2,p}$ -TAF-MSC [59], T-UMC [60] and Orth-NTF [61]. Specifically,

TABLE III
PARAMETERS SELECTION RANGE OF COMPARISON METHODS

Method	Parameter 1	Parameter 2	Parameter 3	Parameter 4	Parameter 5
DiMSC	[0.01, 0.03]	[20, 180]	-	-	-
LT-MSC	[0.01, 100]	-	-	-	-
ETMLSC	[8e-4, 0.01]	-	-	-	-
HLR-M ² VS	[0.01, 0.2]	[0.1, 0.9]	-	-	-
GNLTA	[0.001, 0.1]	[1.1, 2]	-	-	-
MLRR	[0.01, 500]	[0.005, 1]	[1, 5]	-	-
LSGMC	[1e-5, 100]	[0.001, 1]	-	-	-
WTSNM	[0.005, 1]	[1e-8, 10]	(0, 1]	(0, 100]	-
t-SVD-MSC	[0.1, 2]	-	-	-	-
FAMvC	[1e-5, 1e5]	[1e-5, 1e5]	-	-	-
EMKIC	[1, 10]	[1, 10]	[1.3, 6]	[0.1, 1]	[0.001, 0.1]
L1SL	[1e-5, 10]	[1e-5, 10]	[0.9, 1]	-	-
T-UMC	[0.3, 1.2]	[0.3, 1.2]	[0.3, 1.2]	-	-
Orth-NTF	[0.5, 1]	[0.1, 1]	[100, 5000]	-	-

ETLMSC and GNLTA belong to tensor-oriented multi-view clustering methods based on Markov chain. MLRR, LT-MSC, t-SVD-MSC, HLR-M² VS and T-UMC are MVSC methods inherited from LRR. DiMSC studies the diversity of multiple views using the Hilbert-Schmidt independence criterion. WTSNM, LSGMC, L1SL- $\ell_{2,p}$ -TAF-MSC and Orth-NTF are representative non-convex low-rank regularized multi-view clustering methods.

3) *Evaluation Metrics*: To effectively evaluate the performance of clustering, we use six common metrics [62], that is accuracy (ACC), normalized mutual information (NMI), adjusted rank index (AR), F-score, Precision, and Recall. The higher values indicate better clustering performance for the metrics we used.

4) *Parameter Settings*: The proposed method has five parameters that need to be tuned, namely λ_1 , λ_2 , α , τ and θ . Empirically, we set $\lambda_1 \in \{0.0001, 0.001, 0.01, 0.1, 1, 10, 100\}$, $\lambda_2 \in \{0.00001, 0.0001, 0.001, 0.01, 0.1, 1, 10, 100\}$, $\tau \in [0.2 : 0.01 : 0.3]$, $\theta \in \{0.0001, 0.001, 0.01, 0.1, 1, 10, 100\}$ and fix $\alpha = 10^{-11}$. To ensure the fairness of the experiments, we run source codes provided by the authors using the parameters recommended in the articles. For all compared methods, we clearly present the specific parameter settings in Table III.

5) *Experimental Environment*: All experiments are performed by MATLAB R2020b on a workstation equipped with an Intel(R) Xeon(R) Gold 6230 CPU and 256 GB RAM.

B. Experimental Results and Analysis

The clustering results of our method and all competing methods on nine datasets are presented in Tables IV, V, and VI. The best and second best results are highlighted in **bold** and underline, respectively. From the results shown in these tables, we have the following observations.

From a global view, it can be seen that the proposed method outperforms all compared methods over all metrics and datasets in most cases. It is worth noting that our method achieves the ideal clustering performance on five datasets, including Yale dataset, ORL dataset, 20NewsGroups dataset, COIL20 dataset and UCI-Digits dataset. For instance, on the Yale dataset, our method achieves 100% across all metrics, improving upon the second-best method by 1.5%, 1.1%, 2.7%, 2.5%, 3% and 1.5%

TABLE IV
CLUSTERING RESULTS ON 20NEWSGROUPS, YALE AND ORL DATASETS

Datasets	20Newsgrroups						Yale						ORL					
Methods	ACC	NMI	AR	F-score	Precision	Recall	ACC	NMI	AR	F-score	Precision	Recall	ACC	NMI	AR	F-score	Precision	Recall
DiMSC [23]	0.968	0.913	0.923	0.939	0.839	0.940	0.698	0.711	0.523	0.554	0.538	0.570	0.831	0.929	0.783	0.786	0.746	0.830
LT-MSC [38]	0.988	0.958	0.970	0.976	0.976	0.976	0.756	0.780	0.630	0.638	0.613	0.665	0.804	0.921	0.752	0.758	0.707	0.816
ETLMSC [57]	0.952	0.872	0.885	0.908	0.907	0.909	0.698	0.739	0.562	0.590	0.561	0.624	0.908	0.980	0.906	0.908	0.860	0.963
HLR-M ² VS	0.986	0.953	0.965	0.972	0.972	0.972	0.794	0.782	0.562	0.631	0.607	0.656	1.000	1.000	1.000	1.000	1.000	1.000
GNLTA [22]	0.935	0.847	0.847	0.878	0.874	0.881	0.739	0.763	0.606	0.631	0.609	0.655	0.917	0.982	0.917	0.919	0.877	0.967
MLRR [24]	0.956	0.873	0.893	0.956	0.956	0.966	0.736	0.762	0.482	0.820	0.741	0.930	0.815	0.925	0.739	0.836	0.843	0.987
LSGMC [55]	0.944	0.845	0.866	0.892	0.891	0.894	0.741	0.779	0.609	0.634	0.609	0.661	0.836	0.924	0.772	0.777	0.727	0.836
WTSNM [30]	0.986	0.953	0.965	0.972	0.972	0.972	0.985	0.989	0.973	0.975	0.970	0.985	0.988	0.997	0.987	0.987	0.979	0.996
t-SVD-MSC [27]	1.000	1.000	1.000	1.000	1.000	1.000	0.964	0.953	0.913	0.918	0.910	0.927	0.972	0.994	0.973	0.973	0.957	0.990
FAMvC [56]	0.980	0.934	0.950	0.960	0.960	0.960	0.794	0.790	0.629	0.653	0.626	0.680	0.848	0.929	0.878	0.858	0.757	0.912
EMKIC [58]	0.988	0.960	0.970	0.976	0.976	0.976	0.848	0.898	0.819	0.832	0.746	0.939	0.905	0.954	0.889	0.883	0.821	0.955
L1SL- $\ell_{2,p}$ -TAF-MSC [59]	0.990	0.969	0.975	0.980	0.980	0.980	0.913	0.926	0.862	0.871	0.858	0.884	0.994	0.999	0.994	0.994	0.989	0.998
T-UMC [60]	0.913	0.958	0.857	0.917	0.886	0.883	0.830	0.818	0.696	0.726	0.710	0.691	0.913	0.958	0.857	0.917	0.886	0.883
Orth-NTF [61]	0.964	0.909	0.913	0.930	0.929	0.931	0.903	0.910	0.827	0.837	0.832	0.842	0.815	0.890	0.710	0.716	0.698	0.736
UFCTNM	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

TABLE V
CLUSTERING RESULTS ON NOTTING-HILL, UCI-DIGITS AND MITINDOOR DATASETS

Datasets	Notting-Hill						UCI-Digits						MITIndoor					
Methods	ACC	NMI	AR	F-score	Precision	Recall	ACC	NMI	AR	F-score	Precision	Recall	ACC	NMI	AR	F-score	Precision	Recall
DiMSC [23]	0.837	0.799	0.787	0.834	0.822	0.827	0.703	0.772	0.652	0.695	0.673	0.718	0.246	0.383	0.128	0.141	0.138	0.144
LT-MSC [38]	0.907	0.799	0.796	0.840	0.849	0.832	0.924	0.858	0.841	0.857	0.856	0.859	0.416	0.541	0.274	0.285	0.278	0.292
ETLMSC [57]	0.939	0.893	0.874	0.901	0.921	0.883	0.999	0.999	0.999	0.999	0.999	0.999	0.775	0.899	0.729	0.733	0.709	0.758
HLR-M ² VS [28]	0.985	0.956	0.965	0.973	0.977	0.968	0.997	0.999	0.998	0.998	0.998	0.998	0.831	0.917	0.805	0.808	0.770	0.850
GNLTA [22]	0.911	0.853	0.833	0.844	0.844	0.844	0.984	0.981	0.988	0.983	0.981	0.977	0.806	0.915	0.762	0.766	0.746	0.788
MLRR [24]	0.972	0.938	0.941	0.971	0.972	0.979	0.981	0.954	0.958	0.981	0.981	0.993	0.270	0.417	0.135	0.291	0.294	0.974
LSGMC [55]	0.852	0.711	0.689	0.758	0.750	0.766	0.971	0.936	0.937	0.943	0.943	0.943	0.436	0.566	0.298	0.309	0.275	0.353
WTSNM [30]	0.994	0.978	0.987	0.989	0.991	0.987	0.997	0.992	0.993	0.994	0.994	0.994	0.757	0.809	0.647	0.653	0.637	0.669
t-SVD-MSC [27]	0.955	0.895	0.894	0.917	0.934	0.900	0.999	0.996	0.997	0.997	0.997	0.997	0.684	0.750	0.555	0.562	0.543	0.582
FAMvC [56]	0.755	0.639	0.654	0.730	0.730	0.730	0.877	0.879	0.829	0.846	0.841	0.851	0.233	0.330	0.079	0.101	0.063	0.245
EMKIC [58]	0.786	0.684	0.693	0.644	0.642	0.647	1.000	1.000	1.000	1.000	1.000	1.000	0.811	0.910	0.783	0.793	0.700	0.915
L1SL- $\ell_{2,p}$ -TAF-MSC [59]	0.988	0.967	0.973	0.979	0.985	0.973	0.998	0.995	0.996	0.996	0.996	0.996	0.960	0.988	0.957	0.958	0.936	0.980
T-UMC [60]	0.859	0.728	0.758	0.744	0.751	0.682	0.994	0.985	0.987	0.987	0.987	0.986	0.700	0.776	0.605	0.667	0.634	0.614
Orth-NTF [61]	0.778	0.672	0.631	0.715	0.693	0.738	0.991	0.979	0.980	0.982	0.982	0.982	0.750	0.875	0.661	0.667	0.651	0.683
UFCTNM	0.998	0.991	0.995	0.996	0.997	0.995	1.000	1.000	1.000	1.000	1.000	1.000	0.974	0.989	0.969	0.969	0.957	0.981

TABLE VI
CLUSTERING RESULTS ON ALOI, COIL20 AND 100LEAVES DATASETS

Datasets	ALOI						COIL20						100leaves					
Methods	ACC	NMI	AR	F-score	Precision	Recall	ACC	NMI	AR	F-score	Precision	Recall	ACC	NMI	AR	F-score	Precision	Recall
DiMSC [23]	0.522	0.487	0.260	0.345	0.297	0.410	0.796	0.849	0.741	0.755	0.745	0.765	0.524	0.745	0.376	0.382	0.363	0.403
LT-MSC [38]	0.948	0.896	0.888	0.899	0.898	0.900	0.762	0.846	0.688	0.706	0.679	0.732	0.745	0.871	0.634	0.638	0.693	0.677
ETLMSC [57]	0.883	0.919	0.858	0.873	0.828	0.926	0.902	0.958	0.890	0.896	0.869	0.925	0.884	0.973	0.879	0.880	0.826	0.942
HLR-M ² VS [28]	0.966	0.932	0.926	0.933	0.932	0.934	0.833	0.945	0.818	0.828	0.759	0.913	0.621	0.784	0.474	0.479	0.459	0.502
GNLTA [22]	0.880	0.865	0.807	0.827	0.806	0.850	0.840	0.900	0.816	0.825	0.821	0.829	0.921	0.976	0.910	0.911	0.881	0.943
MLRR [24]	0.929	0.879	0.851	0.929	0.929	0.973	0.803	0.927	0.794	0.850	0.843	0.979	0.811	0.926	0.716	0.831	0.828	0.994
LSGMC [55]	0.618	0.608	0.435	0.499	0.441	0.573	1.000	1.000	1.000	1.000	1.000	1.000	0.531	0.750	0.384	0.390	0.353	0.436
WTSNM [30]	0.938	0.883	0.866	0.880	0.877	0.883	0.994	0.999	0.994	0.994	0.989	0.998	0.949	0.989	0.944	0.945	0.916	0.976
t-SVD-MSC [27]	0.957	0.916	0.908	0.917	0.916	0.918	0.809	0.887	0.964	0.776	0.758	0.796	0.942	0.985	0.935	0.936	0.911	0.962
FAMvC [56]	0.841	0.758	0.679	0.712	0.692	0.733	0.768	0.898	0.750	0.764	0.661	0.904	0.924	0.956	0.883	0.884	0.873	0.895
EMKIC [58]	0.983	0.966	0.963	0.967	0.967	0.967	0.921	0.937	0.883	0.896	0.854	0.942	0.912	0.964	0.890	0.892	0.849	0.939
L1SL- $\ell_{2,p}$ -TAF-MSC [59]	0.980	0.959	0.955	0.960	0.960	0.960	0.973	0.979	0.953	0.955	0.946	0.964	0.961	0.992	0.960	0.960	0.935	0.986
T-UMC [60]	0.823	0.847	0.765	0.891	0.823	0.802	0.700	0.776	0.605	0.667	0.634	0.614	0.602	0.780	0.447	0.489	0.467	0.462
Orth-NTF [61]	0.852	0.821	0.728	0.755	0.747	0.763	0.853	0.867	0.744	0.757	0.746	0.768	0.704	0.882	0.622	0.625	0.609	0.642
UFCTNM	0.985	0.970	0.968	0.971	0.971	0.971	1.000	1.000	1.000	1.000	1.000	1.000	0.971	0.992	0.967	0.967	0.952	0.983

in the six evaluated metrics. Although competing methods such as HLR-M² VS, EMKIC, t-SVD-MSC and LSGMC also get ideal results on some datasets, our proposed method has an overwhelming advantage over all experimental results. Moreover, on the remaining four datasets, the improvement of our proposed method is considerably remarkable. For example, on the MITIndoor dataset, our method outperforms the suboptimal L1SL- $\ell_{2,p}$ -TAF-MSC by 1.4%, 0.1%, 1.2%, 1.1%, 2.1%, 0.1%, respectively. These observations consistently verify the effectiveness of our proposed method. In general, the proposed method can always achieve satisfactory performance in different applications, which further validates its robustness for complex clustering scenarios.

Next, we provide fine-grained analyses of the experimental results from a local perspective.

- 1) In most cases, tensor-oriented multi-view clustering methods are superior than the matrix-oriented ones. The main reason is that compared with matrix-oriented methods, tensor-oriented methods can capture high-order correlations among multiple views, which indicates that the learned affinity matrix can fully utilize the complementary information and high-order information embedded in the multi-view data, thus a more comprehensive tensor representation can be learned.
- 2) Although most tensor representation based methods have achieved impressive performance, there are still some

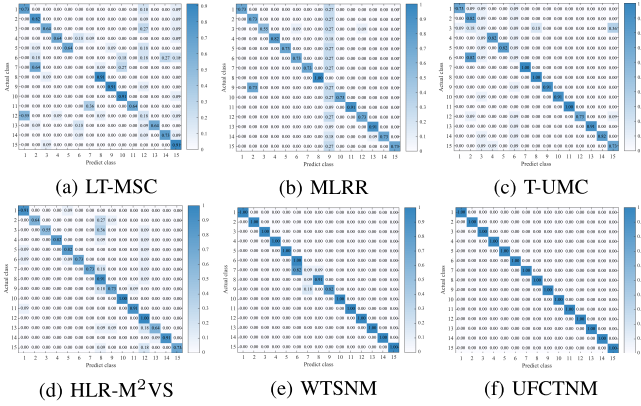


Fig. 3. The confusion matrices comparison among six methods on the Yale dataset.

notable differences. First, t-SVD based methods are significantly better than other methods. Since t-SVD treats a third-order tensor as linear operator on matrices, which can effectively characterize the intrinsic low-rank structure of the tensor and ensure consensus among multiple views, while avoiding inherent information loss. Second, the non-convex low-rank regularized methods are always superior to convex low-rank regularized ones, especially compared to the important baseline t-SVD-MSC. For example, on the Notting-Hill dataset, WTSNM outperforms t-SVD-MSC with significant improvements over six metrics, i.e., 3.9%, 8.3%, 9.3%, 7.2%, 5.7%, 8.7%. This indicates that non-convex low-rank regularized method is more capable of ensuring the high-order consistency and fully capturing the complementarity among multi-views.

- 3) Most importantly, our method is always superior to other non-convex low-rank regularized methods. On the MITIndoor dataset, our method obtains the best performance and outperforms non-convex methods L1SL- $\ell_{2,p}$ -TAF-MSC, GNLTa and WTSNM with significant improvements over ACC value, i.e., 2.4%, 16.8%, 21.7%, respectively. This is mainly attributed to the fact that our method can better capture the intrinsic low-rank structure of high-dimensional data and suppress the influence of outliers. Moreover, the utilization of hyper-Laplacian and spectral embedding enables our method to preserve view-specific local similarity information and encode discriminative information in representations.

The above observations verify the effectiveness of our method, which provides experimental evidences that our proposed method is an effective and robust MVSC method.

C. Results Visualization

To further substantiate the effectiveness of our method, we provide confusion matrix visualizations on the Yale dataset, facilitating a comparative analysis against LT-MSC, MLRR, T-UMC, HLR-M²VS, WTSNM, and our proposed method. As shown in Fig. 3, the row and column correspond to the predicted labels and the ground-truth labels, respectively. Here, the predicted labels are calculated by performing a permutation

mapping function in ACC. The i -th diagonal block represents the observations that are classified correctly, while the (i, j) -th ($i \neq j$) block denotes the incorrectly classified ones. We can see that the confusion matrices in Fig. 3(b)–(e) have significant confusion in some subjects and their non-diagonal elements are larger than those of our method, resulting in lower clustering accuracy. In particular, the accuracy of the LT-MSC method in subjects 6 and 7 is 0, which seriously affects the overall clustering results. Our method shows notably superior performance in terms of accuracy across all categories compared to other methods. Specifically, the diagonal elements of Fig. 3(f) are all equal to 1, indicating that our method makes almost no misclassifications. The reason is that the learned tensor representation in our method encodes the cluster structure well.

A high-quality affinity matrix is the key for spectral clustering. The visualizations of affinity graphs generated by DiMSC, LT-MSC, t-SVD-MSC, WTSNM, T-UMC, MLRR and the proposed method on ORL and COIL20 datasets are shown in Fig. 4. For the ORL dataset, it can be intuitively observed that the block diagonal structure of the affinity matrices in Fig. 4(a)–(f) are indistinct and contain substantial noise. In particular, the block diagonal structure of the affinity graphs learned by DiMSC, t-SVD-MSC and WTSNM is obviously more blurred than that of our method. Our method outperforms other methods in revealing the intrinsic clustering structure of data. Specifically, the affinity matrix in Fig. 4(g) exhibits a clearer block diagonal structure, while the non-block diagonal elements are generally smaller, which means that our method has less deceptive pairwise correlations. The similar conclusion can be drawn from Fig. 4 for the ORL dataset. Thus, our affinity graph can characterize the cluster structure well and provide more discriminative subspace information for the clustering process.

Furthermore, we perform a two-dimensional visualization to compare the low rankness of coefficient matrices learned by our method and other competing methods. Specifically, the visualization can be plotted by the t -distributed stochastic neighbor embedding (t -SNE) [63] technique with the input cluster indicator matrix F . The embedding results of 20Newsgroup and COIL20 datasets are shown in Fig. 5. It can be seen that the samples in Fig. 5(a)–(e) are mixed and difficult to classify correctly. In particular, visualization results obtained by DiMSC, HLR-M²VS and GNLTa exist small gaps between clusters dyed with different colors, while points within the same cluster are farther to each other. Our method is completely correct in the cluster division of datasets as shown in Fig. 5(f), which consistently agrees with the clustering results in Tables IV and VI. This means that the learned coefficient matrices have a more compact subspace structure, which better reflects its superiority in maintaining complementarity.

D. Ablation Studies

In the proposed model (16), the hyper-Laplacian regularization and spectral embedding are used simultaneously. To systematically evaluate their contributions, comprehensive ablation studies are carried out. We consider the following four cases for experiments. In case I, the (13) is used as the basic model, which

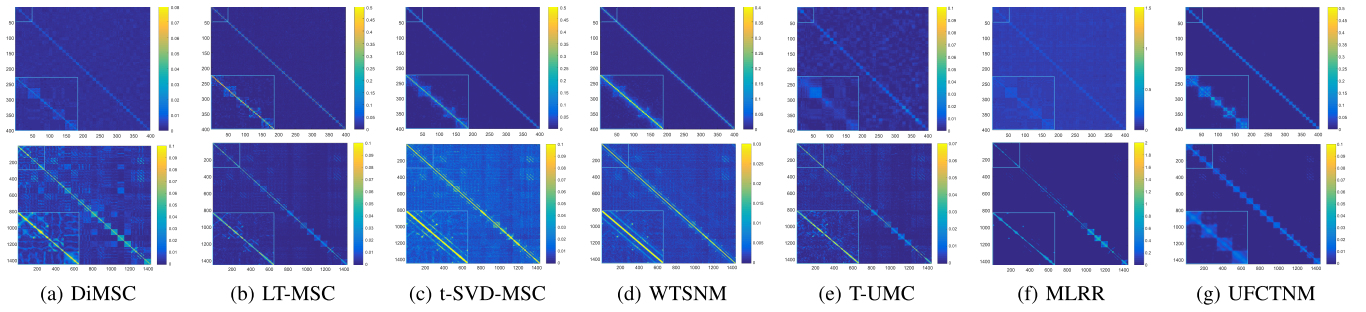


Fig. 4. Comparison of affinity matrices of different methods with our method. The first row are the results on the ORL dataset, and the second row are the results on the COIL20 dataset.

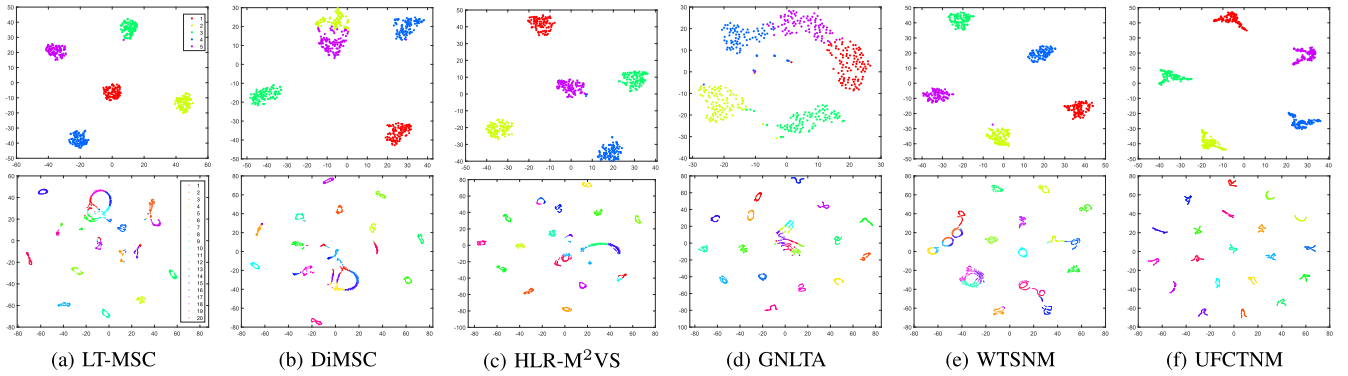


Fig. 5. Visualization results of different methods. The first row are the results on the 20News groups, and the second row are the results on the COIL20.

TABLE VII
ABLATION STUDY OF THE PROPOSED UFCTNM

Datasets	T ₁	T ₂	ACC	NMI	F-score	Precision	Recall
100leaves	✗	✗	0.955	0.990	0.954	0.934	0.975
	✓	✗	0.961	0.991	0.957	0.936	0.979
	✗	✓	0.961	0.991	0.957	0.934	0.981
	✓	✓	0.971	0.992	0.967	0.952	0.983
ALOI	✗	✗	0.960	0.921	0.922	0.921	0.923
	✓	✗	0.981	0.963	0.964	0.964	0.964
	✗	✓	0.980	0.961	0.962	0.961	0.964
	✓	✓	0.985	0.970	0.971	0.971	0.971
Notting-Hill	✗	✗	0.992	0.973	0.987	0.989	0.985
	✓	✗	0.995	0.981	0.990	0.992	0.988
	✗	✓	0.993	0.973	0.987	0.990	0.985
	✓	✓	0.998	0.991	0.996	0.997	0.997
MITIndoor	✗	✗	0.931	0.981	0.934	0.908	0.963
	✓	✗	0.936	0.981	0.934	0.901	0.968
	✗	✓	0.938	0.976	0.936	0.907	0.967
	✓	✓	0.974	0.989	0.969	0.957	0.981

includes the low-rank regularizer and the error term. In case II, the hyper-Laplacian regularization is added to (13), which is denoted as T₁. In case III, the spectral embedding is added to (13), which is denoted as T₂. In case IV, we perform experiments using the proposed model (16). The quantitative evaluations across five clustering metrics (ACC, NMI, F-Score, Precision and Recall) are presented in Table VII, with comparative ACC visualizations shown in Fig. 6.

Therefore, we have the following observations.

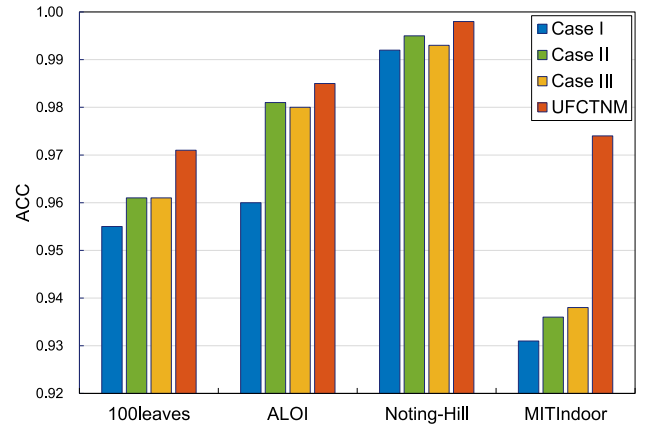


Fig. 6. Comparison with different regularizations on four datasets in terms of ACC.

- 1) The utilization of hyper-Laplacian regularization and spectral embedding both improve the clustering performance, respectively. By comparing case I and case II, we can see that case II gains more than 2% improvement on ALOI dataset in terms of ACC, which validates the effectiveness of hyper-Laplacian regularization to preserve view-specific local geometries. In addition, observed from the results derived from case I and case III, the case III achieves 0.6% and 2% increase over 100leaves and ALOI, respectively, compared to case I, which illustrates the superiority of spectral embedding in significantly

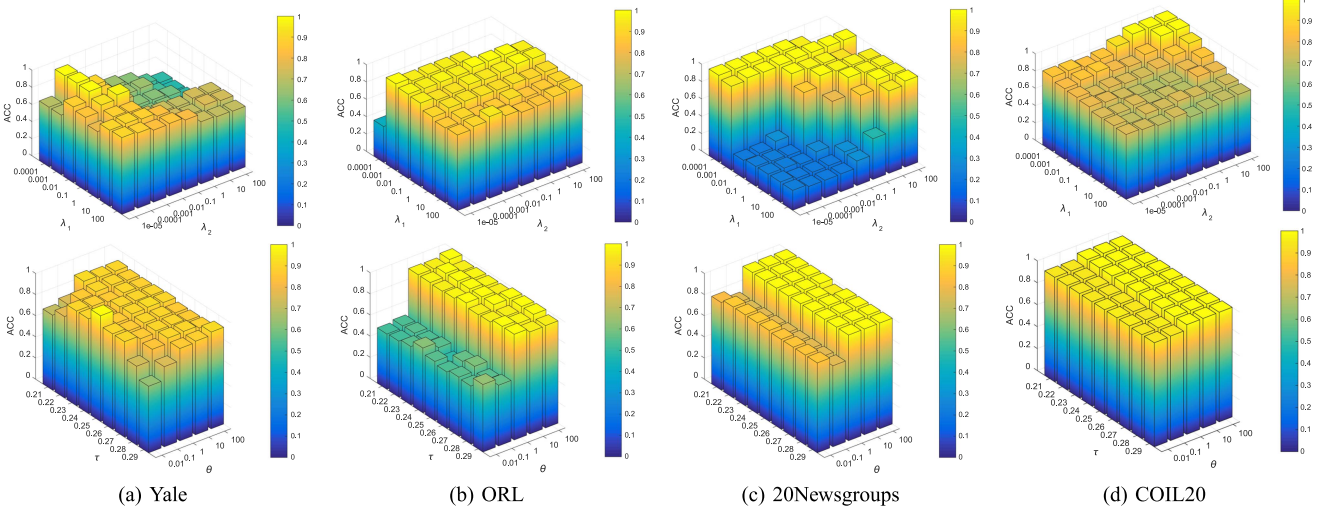


Fig. 7. Sensitivity analysis of λ_1 , λ_2 , τ and θ in terms of ACC over Yale, ORL, 20Newsgrroups and COIL20.

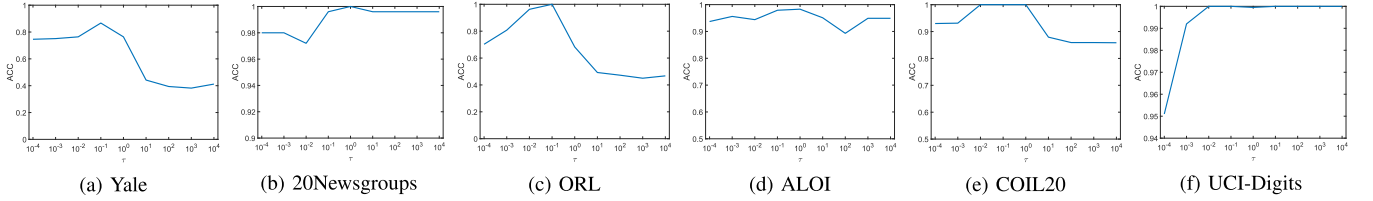


Fig. 8. The ACC with different τ for Yale, 20Newsgrroups, ORL, ALOI, COIL20 and UCI-digits datasets.

improving subspace discriminative ability. As observed from the results of case II and case III, case II achieves better results on ALOI and Notting-Hill datasets, while case III performs better on others. This indicates that the two cases exhibit their respective strengths in different scenarios. Therefore, it is necessary to integrate the two modules into a unified framework to improve clustering performance.

- 2) The proposed complete model (16) achieves notably superior performance on four datasets compared to other cases. This proves the effectiveness of unifying representation learning, hyper-Laplacian regularization and spectral embedding into single framework for MVSC, which emphasizes successful collaboration of these modules, allowing them to exploit the information embedded in multi-view data and achieve commendable performance.

Overall, the proposed method integrates two regularizers into a unified framework to preserve the global low-rank structure and view-specific local similarity information simultaneously, as well as encode discriminative information in representations.

E. Parameter Sensitivity Analysis

To explore the sensitivity of the proposed method to parameter variations, we investigate the impact of four parameters λ_1 , λ_2 , τ , θ on the clustering performance of four datasets in Fig. 7, which shows how the performance of our method varies under different parameter combinations, highlighting fluctuations within various value ranges. By fixing the values of parameters

τ , θ and α , the ACC values for different combinations of λ_1 and λ_2 are shown in the first row of Fig. 7. We can observe that when λ_1 takes $\{0.001, 0.001, 0.01, 0.1, 1, 10, 100\}$ and λ_2 in $\{0.0001, 0.001, 0.01, 0.1, 1, 10, 100\}$, the ACC on the ORL dataset maintains a high level, indicating that the performance of our method is stable across a wide range of parameters. The second row in Fig. 7 examines the influence of θ and τ on ACC when other parameter values are fixed. It is worth noting that the ACC metric stably keeps at a relatively higher score when τ in $[0.21 : 0.01 : 0.29]$ and θ takes $\{1, 10, 100\}$, which implies that our method has robust capability to cope with clustering perturbations. Moreover, we find the optimal range of parameter τ with parameter tuning via warm start strategy. By fixing the values of other parameters, the results generated on six datasets are reported in Fig. 8. It can be seen that the clustering performance of the proposed achieves optimal on all six datasets when τ is in the interval $[0.1, 1]$.

Therefore, these results demonstrate that satisfactory performance can be achieved through parameter tuning, and the stability of parameters also directly illustrates the robustness and efficiency of our method.

F. Numerical Convergence

To further verify the stability of our method, we investigate the numerical evidence of convergence of the proposed optimization algorithm. Specifically, the relative errors, ACC and NMI values versus iterations on the four datasets are shown in Fig. 9. As shown in Fig. 9, as the number of iterations increases, the ACC

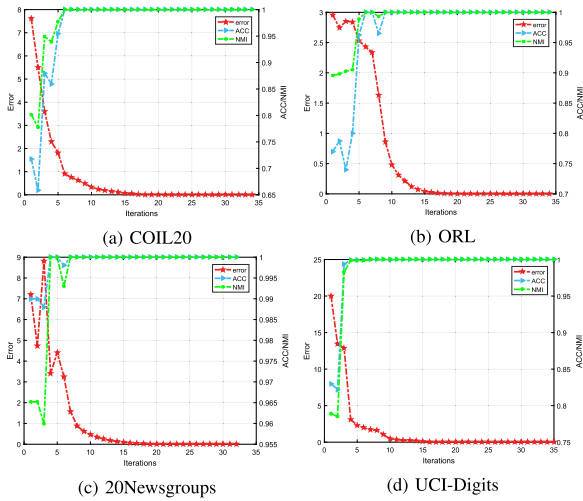


Fig. 9. Relative errors, ACC, and NMI curves on the COIL20, ORL, 20Newsgroups and UCI-Digits datasets, respectively.

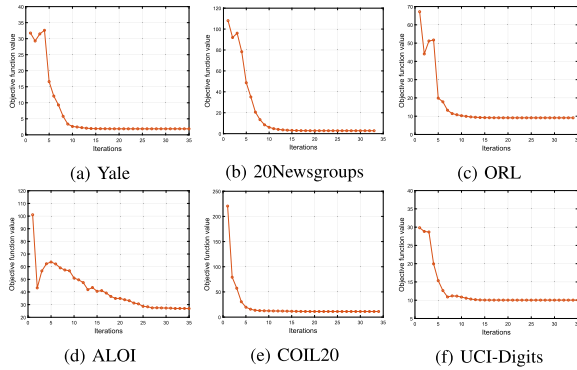


Fig. 10. The objective function value versus the number of iterations of UFCTNM on six datasets, respectively.

and NMI values tend to be stable and reach the optimal value about ten iterations, and the relative errors rapidly produce stable values. These results indicate that the proposed algorithm can achieve convergence reliably and efficiently. Furthermore, we visualize the objective function values versus the number of iterations of our proposed algorithm on the Yale, 20Newsgroups, ORL, ALOI, COIL20 and UCI-digits datasets. As shown in Fig. 10, the objective function value gradually descends to a stable point in about 25 iterations in most cases. Despite minimal fluctuations, the objective function values on six datasets decrease and eventually reach a stationary state, further demonstrating the stability and convergence of our algorithm.

Thus, the convergence of our algorithm can not only be theoretically proved, but also be experimentally verified.

V. CONCLUSION

In this paper, we proposed a novel MVSC model called UFCTNM by employing non-convex regularization. On one hand, it utilizes CTNN to effectively capture the global low-rank structure in multi-view data and enhances the robustness to outliers by introducing capped $\ell_{2,1}$ norm. On the other hand,

it integrates hyper-Laplacian regularization with spectral embedding through joint optimization, achieving a mutual boost between geometric structure preservation and discriminative representation learning. Later, through the theoretical analysis of the CTNN regularized least squares subproblem, an efficient optimization method with convergence guarantee is proposed to solve the proposed model. More importantly, despite the inherent non-convexity of the model, its convergence properties were rigorously investigated and established. Finally, the experimental results demonstrated that the proposed model achieves significant performance improvements over existing state-of-the-art models.

While our proposed method achieves superior clustering performance, it is worth noting that Fourier transformation, t-SVD and matrix inversion result in high time complexity. In the future, efficient technologies will be considered to accelerate the algorithm speed. Moreover, the techniques such as adaptive view weighting or view selection may be considered to address issues where certain views significantly contradict others, and we will explore extensions to handle incomplete multi-view settings.

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